VRF Installation Quick Guide

A quick reference guide to all aspects of Midea VRF installation

For in-depth information please always refer to the Midea manuals supplied with the equipment

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Basic System Design

- Pages 4-5 Capacity/Permitted no. of indoors
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Capacity/Permitted No. of IDUs

Mini VRF & Sideflow

Outdoor Unit	Capacity	Max No. Indoors	Sum Capacity Ratio
8	2.5	4	45%-130%
10.5	3	5	45%-130%
12	4	6	45%-130%
14	5	6	45%-130%
16	6	7	45%-130%
18	6.5	9	45%-130%
20	7	10	45%-130%
22.4	7.5	11	45%-130%
26	8	12	45%-130%
40	14	14	45%-130%
45	15	15	45%-130%

Capacity/Permitted No. of IDUs

V4+R Heat Recovery & V6 Heat Pump

Outdoor Unit	Capacity	Max No. Indoors	Sum Capacity Ratio
25.2	8	13	50%-130%
28	10	16	50%-130%
33.5	12	20	50%-130%
40	14	23	50%-130%
45	16	26	50%-130%
53.2	18	29	50%-130%
56	20	33	50%-130%
61.5	22	36	50%-130%
68	24	39	50%-130%
73	26	43	50%-130%
80	28	46	50%-130%
85	30	50	50%-130%
90	32	53	50%-130%
96	34	56	50%-130%
101	36	59	50%-130%
106.5	38	63	50%-130%
113	40	64	50%-130%
……	…….	……	
180	64	64	50%-130%

Piping Distances

Mini VRF

			Permitted Value	Piping
Pipe Length	Total Actual Length		≤ 100m	Refer to Installation Guide for full design
	Max Piping	Actual	≤ 45m (8kw, 10.5kw) ≤ 60m (12kw, 14kw, 16kw, 18kw)	
		Equivalent	≤ 50m (8kw, 10.5kw) ≤ 70m (12kw, 14kw, 16kw, 18kw)	
	First Branch to Furthest IDU		≤ 20m	
	Nearest Branch		≤ 15m	
Drop Height	IDU-ODU Height Difference	ODU Up	≤ 30m	
		ODU Down	≤ 20m	
	IDU-IDU Drop height		≤ 8m	

NOTE: When the total equivalent length of liquid+gas lines is ≥ 90m, the diameter of the gas line must be increased

Piping Distances

Sideflow

			20-26kw	40-45kw	Piping
Pipe Length	Total Actual Length		≤ 120m	≤ 250m	Refer to Installation Guide for full design
	Max Piping	Actual	≤ 60m	≤ 100m	
		Equivalent	≤ 70m	≤ 120m	
	First Branch to Furthest IDU		≤ 20m	≤ 40m	
	Nearest Branch		≤ 15m	≤ 15m	
Drop Height	IDU-ODU Height Differen ce	ODU Up	≤ 30m	≤ 30m	
		ODU Down	≤ 20m	≤ 20m	
	IDU-IDU Drop height		≤ 8m	≤ 8m	

NOTE: When the total equivalent length of liquid+gas lines is ≥ 90m, the diameter of the gas line must be increased

Piping Distances

V4+R Heat Recovery

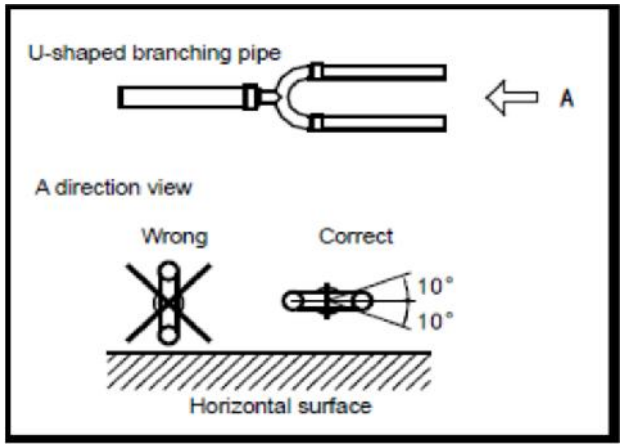
			40-45kw	Piping
Pipe Length	Total Actual Length		≤ 1000m	Refer to Installation Guide for full design
	Max Piping	Actual	≤ 175m	
		Equivalent	≤ 200m	
	First Branch to Furthest IDU		≤ 40/90m	
	MS Box to IDU		≤ 40m	
Drop Height	IDU-ODU Height Difference	ODU Up	≤ 70m	
		ODU Down	≤ 110m	
	IDU-IDU Drop height		≤ 30m	

Piping Distances

V6 Heat Pump

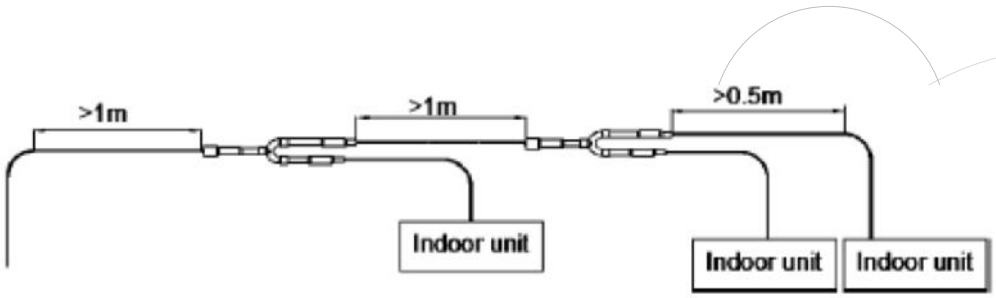
			40-45kw	Piping
Pipe Length	Total Actual Length		≤ 1000m	Refer to Installation Guide for full design
	Max Piping	Actual	≤ 175m	
		Equivalent	≤ 200m	
	First Branch to Furthest IDU		≤ 40/90m	
Drop Height	IDU-ODU Height Difference	ODU Up	≤ 90m	
		ODU Down	≤ 110m	
	IDU-IDU Drop height		≤ 30m	

Piping Instructions



U Shaped INDOOR Ts can be installed horizontal or vertical

If they are installed horizontal they must be within an angle 10°

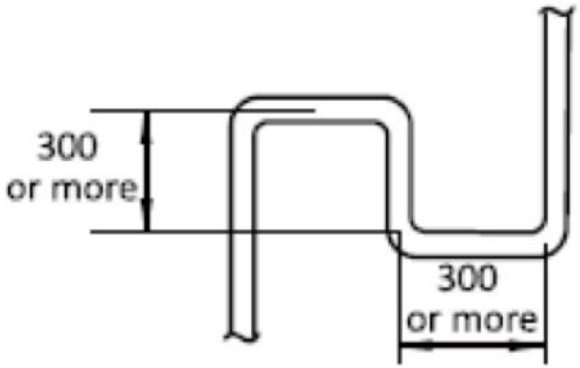


At least 1m between Ts

At least 0.5m must be left between any 90° and

- another 90°
- a MS box
- a fan coil

Piping Instructions



If the outdoor unit is above the fan coils and the difference is greater than 20m, it is recommended that an oil return bend is set every 10m in the suction pipe

If the outdoor unit is below the fan coils and the difference is greater than 40m, the liquid pipe of the main run should be increased to the next size up

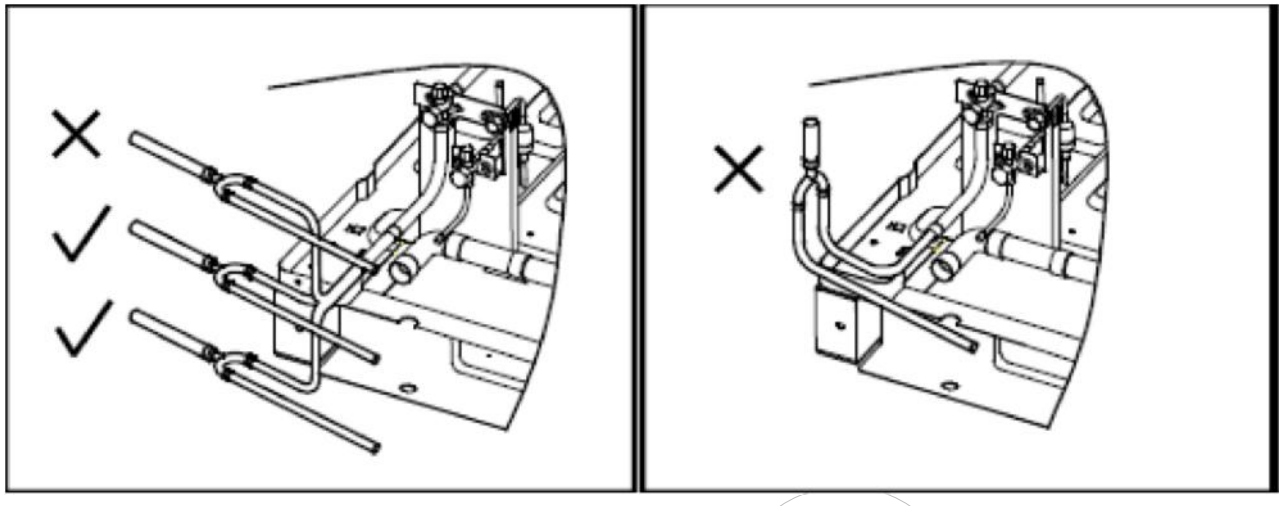
Original	Increased
9.53	12.7
12.7	15.9
15.9	19.1
19.1	22.2
22.2	28.6
28.6	31.8
31.8	38.1

Piping Instructions

MS to IDU The total capacity of Indoor units(×100W)	When indoor unit pipe length≤10m		When indoor unit pipe length>10m	
	Gas side	Liquid side	Gas side	Liquid side
A≤45	Φ12.7mm	Φ6.4mm	Φ15.9mm	Φ9.5mm
A≥56	Φ15.9mm	Φ9.5mm	Φ19.1mm	Φ12.7mm

If the distance from the MS Box/Refnet to the IDU is greater than 10m, both pipe sizes should be increased to the next sizes up. Decrease the sizes back down for the last 0.5m to allow the flare connections

Piping Instructions



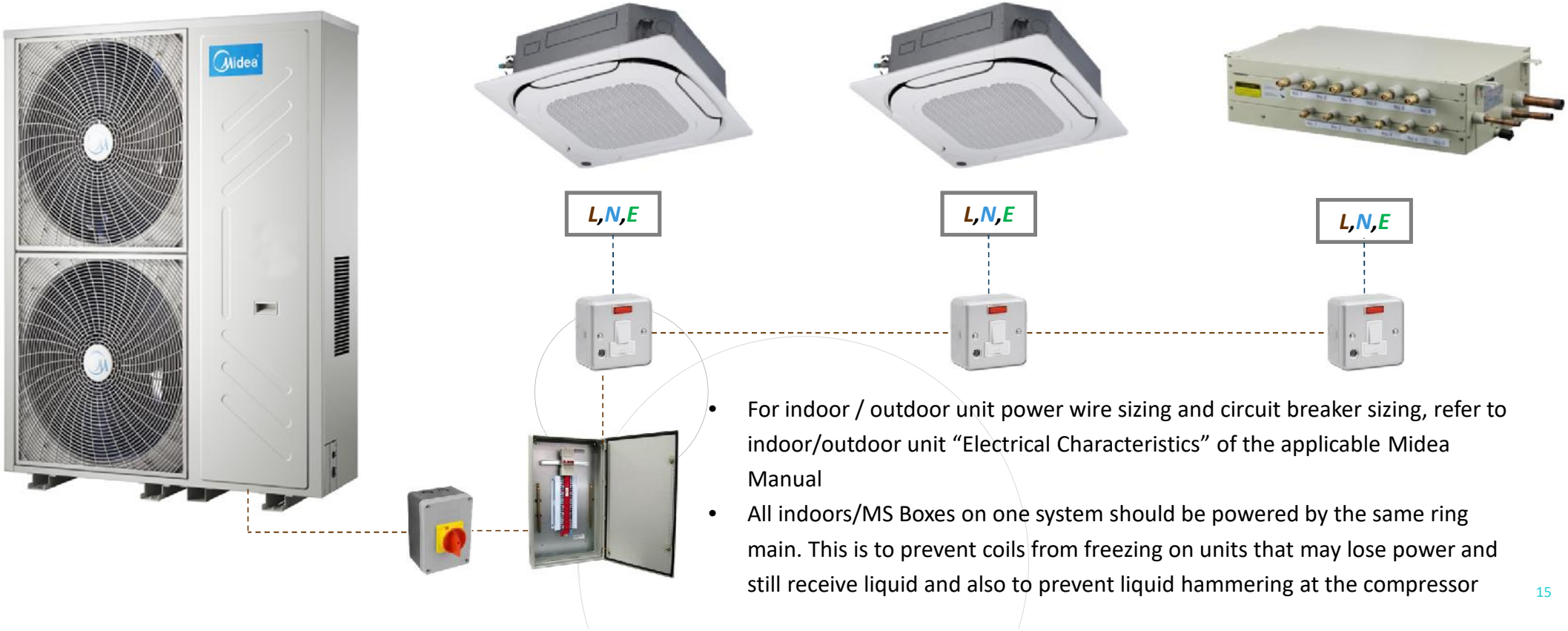
To ensure even distribution of refrigerant and oil return outdoor refnets connecting modular condensers must be:

- Horizontal (within 10°)
- Installed below or level with the service valve

Comms & Power Wiring

- Page 15 Power Wiring
- Page 16 Heat Pump Comms Wiring
- Page 17 Heat Recovery Comms Wiring
- Page 18 Modular Condenser Comms Wiring
- Page 19 Central Controller Wiring

Power Wiring



- For indoor / outdoor unit power wire sizing and circuit breaker sizing, refer to indoor/outdoor unit “Electrical Characteristics” of the applicable Midea Manual
- All indoors/MS Boxes on one system should be powered by the same ring main. This is to prevent coils from freezing on units that may lose power and still receive liquid and also to prevent liquid hammering at the compressor

Communication Wiring

Heat Pump

P,Q,E

P,Q,E

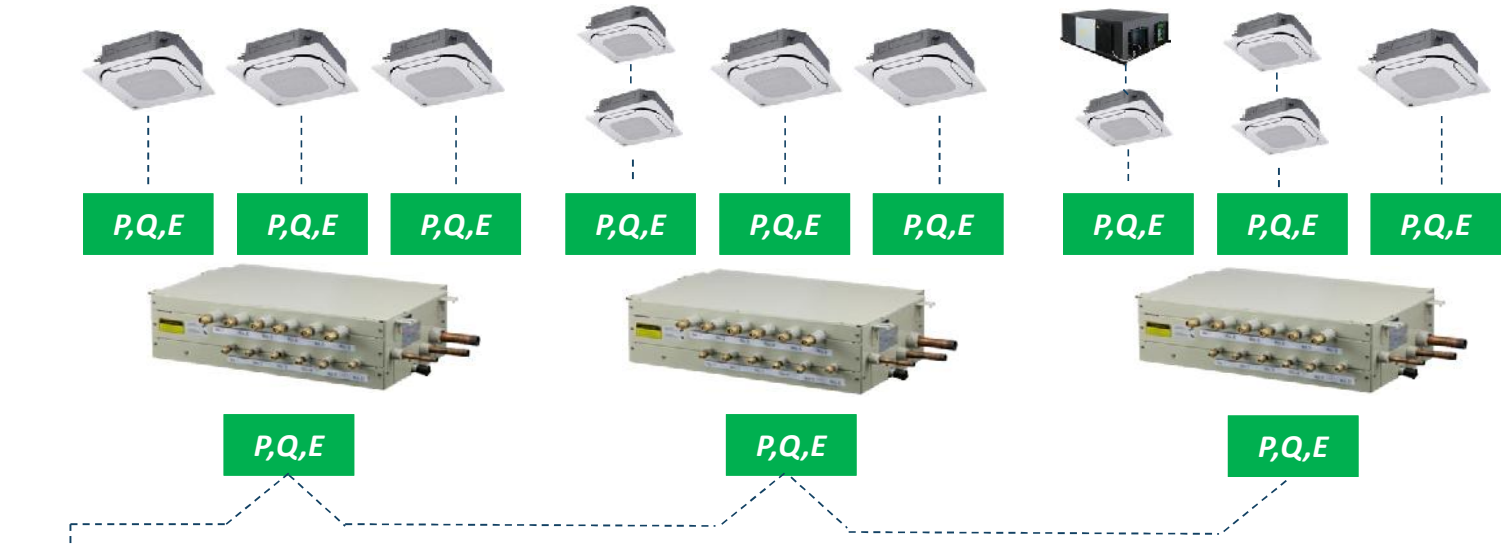
P,Q,E

P,Q,E

- 3 core 0.75mm² shielded cable
- The P Q E communication wires should be connected one unit after another in a daisy chain from the ODU to the final IDU
- At the final unit, a 120Ω resistor (provided in the accessory pack) should be connected between the P and Q terminals
- The shielding nets of the communication wires should be connected together and grounded at one end of the daisy chain
- HRV Ventilation units that are to be controlled via a central controller should be added to the P Q E communication chain

Communication Wiring

Heat Recovery



P,Q,E

- 3 core 0.75mm² shielded cable
- The P Q E communication wires should be connected one unit after another in a daisy chain from the ODU to the final MS Box. These are labelled up as simply P Q E, without any port number reference
- Fan coil units are wired into the P Q E terminals to which they are piped. Each set of piping piping ports (1-12) has its own P Q E terminal for the fan coil(s) connected to it. For multiple fan coils daisy chain the connection. HRV units can be added to any indoor unit P Q E daisy chain
- At the final unit, a 120Ω resistor (provided in the accessory pack) should be connected between the P and Q terminals
- The shielding nets of the communication wires should be connected together and grounded at one end of the daisy chain

Communication Wiring

Modular Condensers

For systems with two or more condensers piped together they are connected on H1 H2 E

- 3 core 0.75mm² shielded cable
- Master slave address settings set via ENC1

Central Controller Wiring

X,Y,E

X,Y,E

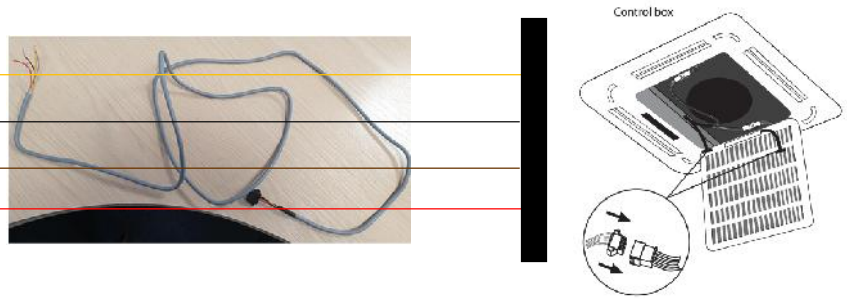
- 3 core 0.75mm² shielded cable
- All models of central controller are wired to the condensing unit on the XYE terminals
- Maximum distance is 200m
- Multiple separate systems are daisy chained together
- Modular condensing units only the master is connected

Controller Wiring

- Page 21-22 KJR29B
- Page 23 WDC86E/KD
- Page 24-27 WDC120G
- Page 28-29 KJR27B

Controller Wiring – KJR29B

1 to 1 Installation

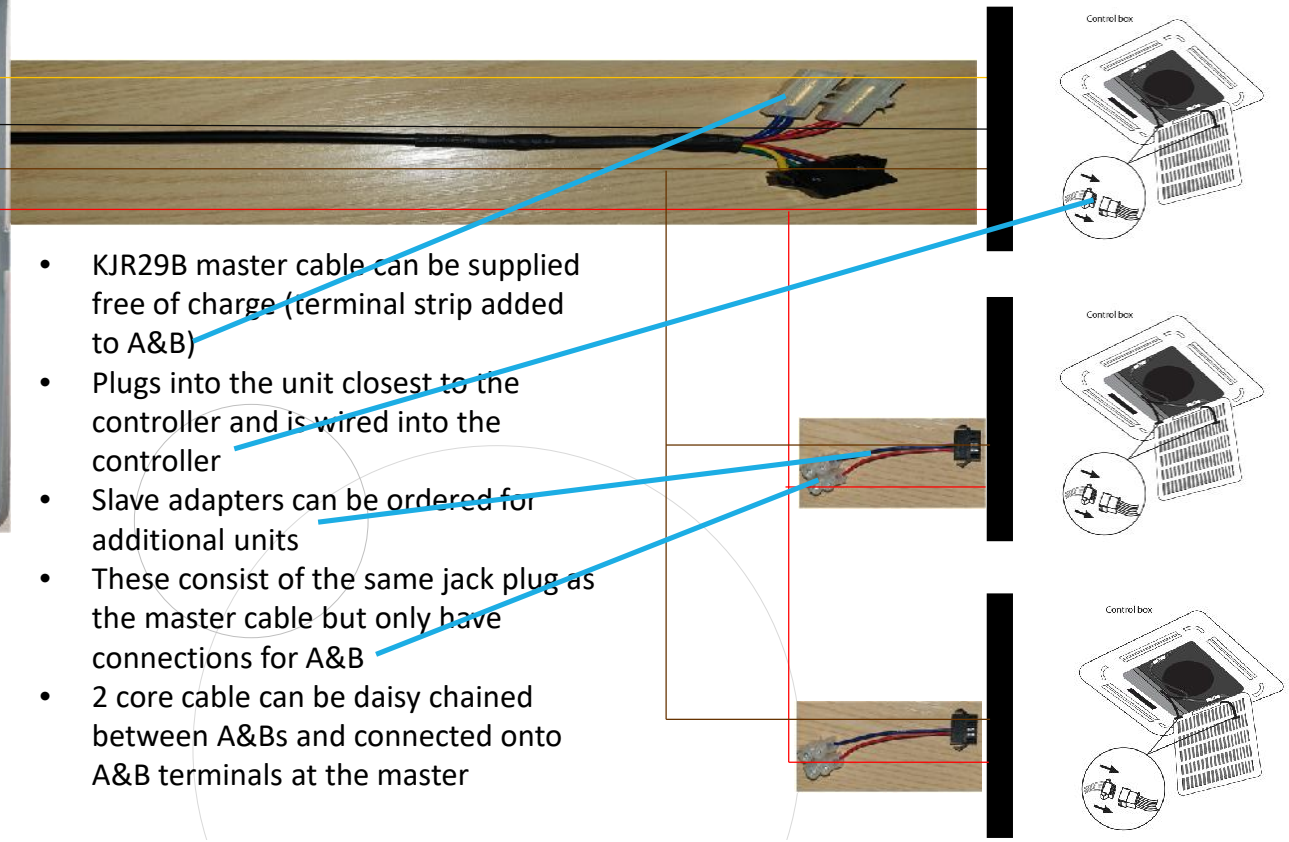


A – Brown/Blue
B – Red
C- Yellow
D - Black

- Comes with a 6m cable
- Jack plugged one side/made off the other
- Jack plugs into fascia/display board
- For ceiling floor units the connection is accessed by removing the right-side unit cover
- Other end is made off and wired into the terminals
- A&B – Comms
- C&D – 5v Power

Controller Wiring – KJR29B

1 to Many Installation

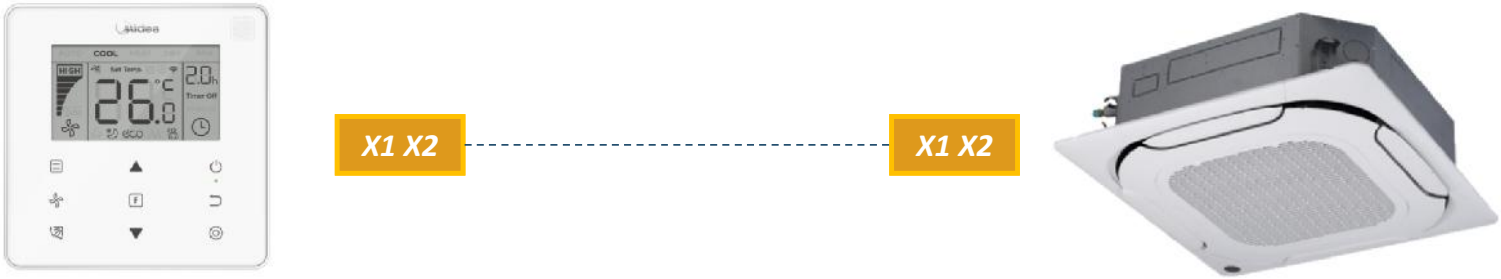


A – Brown/Blue
B – Red
C- Yellow
D - Black

- KJR29B master cable can be supplied free of charge (terminal strip added to A&B)
- Plugs into the unit closest to the controller and is wired into the controller
- Slave adapters can be ordered for additional units
- These consist of the same jack plug as the master cable but only have connections for A&B
- 2 core cable can be daisy chained between A&Bs and connected onto A&B terminals at the master

Controller Wiring – WDC86E/KD

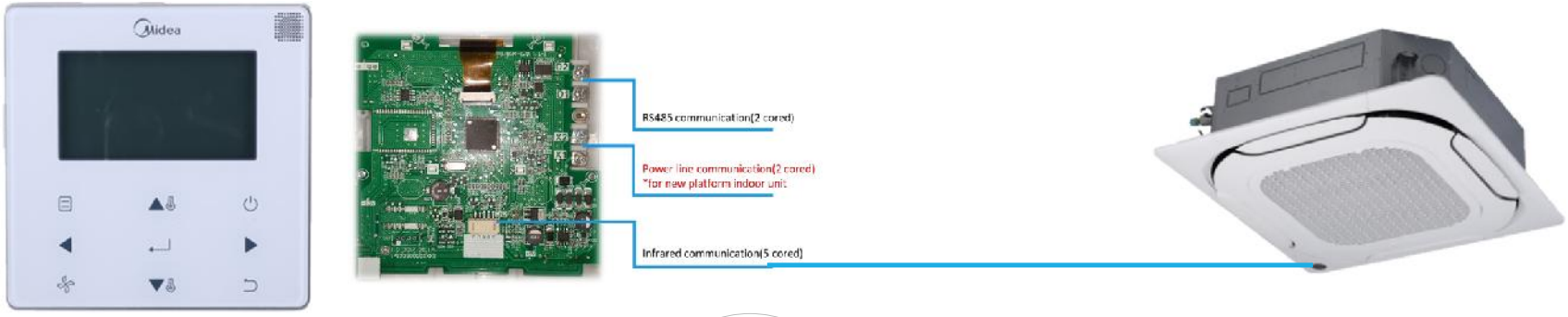
1 to 1 Installation Only



- Controller and fan coil are connected using the X1 X2 terminals
- 2 core shielded cable 0.5mm²-1.5mm²
- Maximum 200m length

Controller Wiring – WDC120G

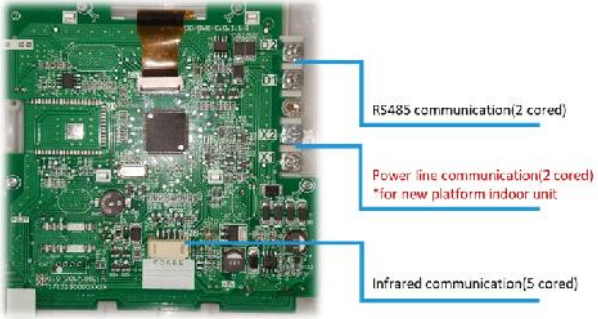
MI Unit - 1 to 1 Installation



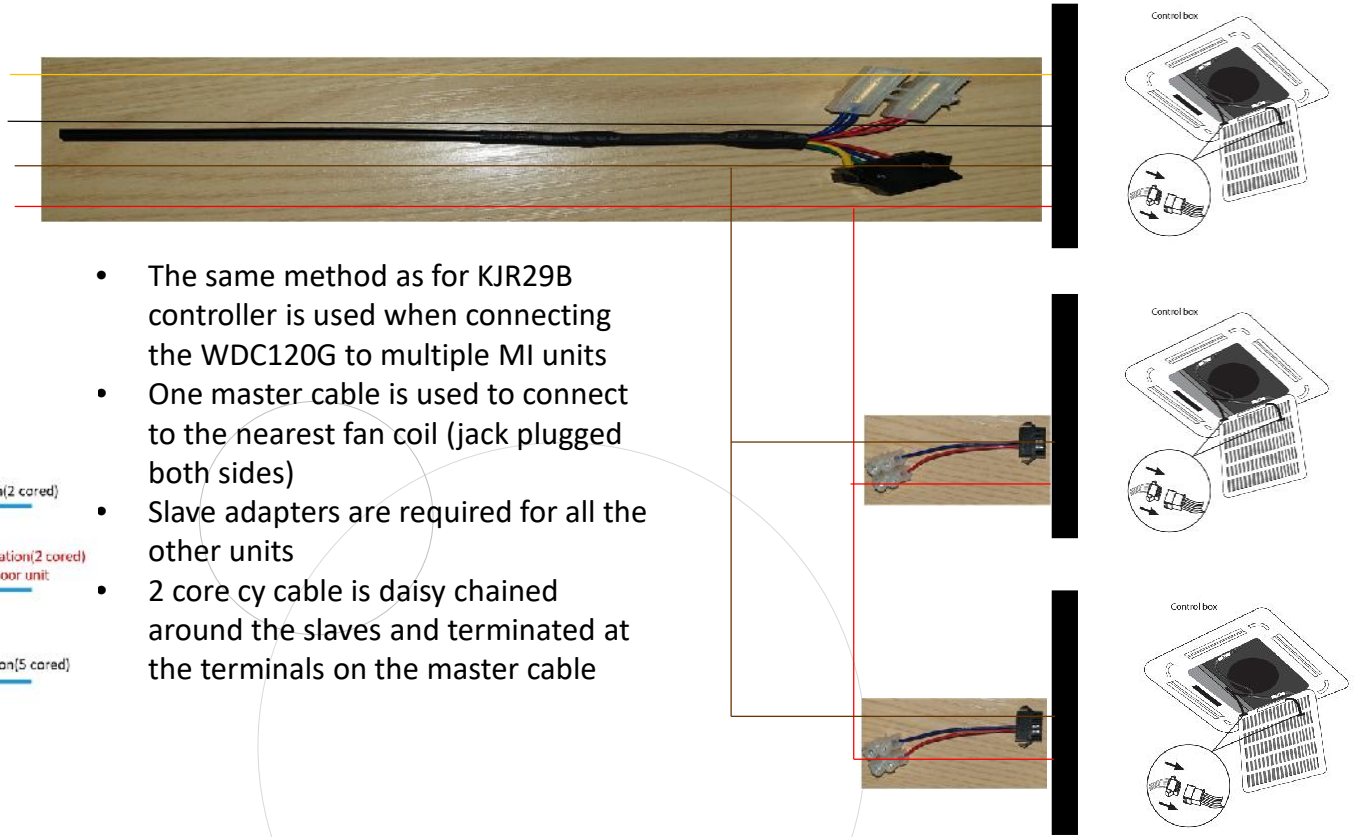
- Controller and fan coil are connected using the the infra red communication sockets on both the controller and the fan coil unit
- The controller comes with a small fly lead cable for temporary connection. As the cable is jack plugged both ends Midea UK will supply a 10m if this method of installation is required on a permanent basis
- Maximum 20m length

Controller Wiring – WDC120G

MI Unit - 1 to Many Installation

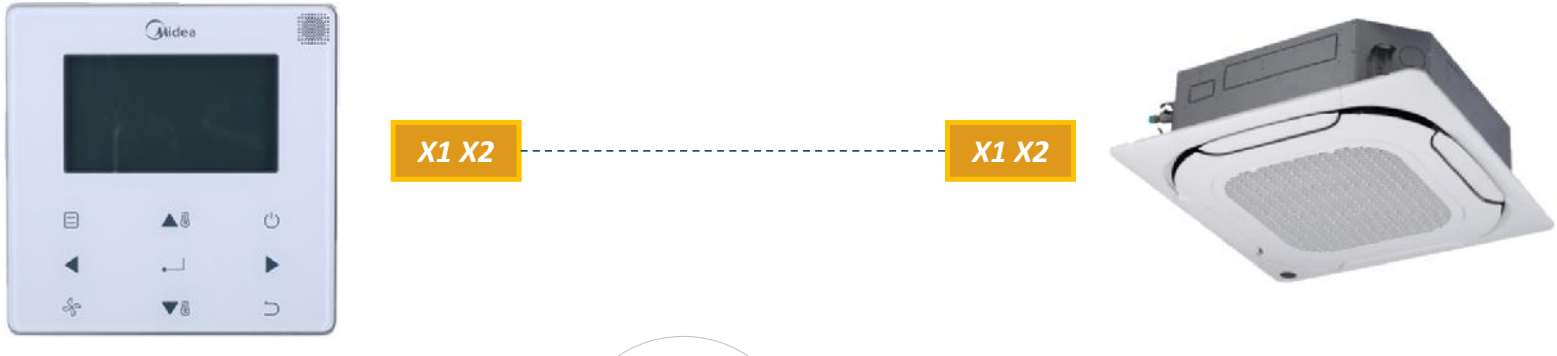


- The same method as for KJR29B controller is used when connecting the WDC120G to multiple MI units
- One master cable is used to connect to the nearest fan coil (jack plugged both sides)
- Slave adapters are required for all the other units
- 2 core cy cable is daisy chained around the slaves and terminated at the terminals on the master cable



Controller Wiring – WDC120G

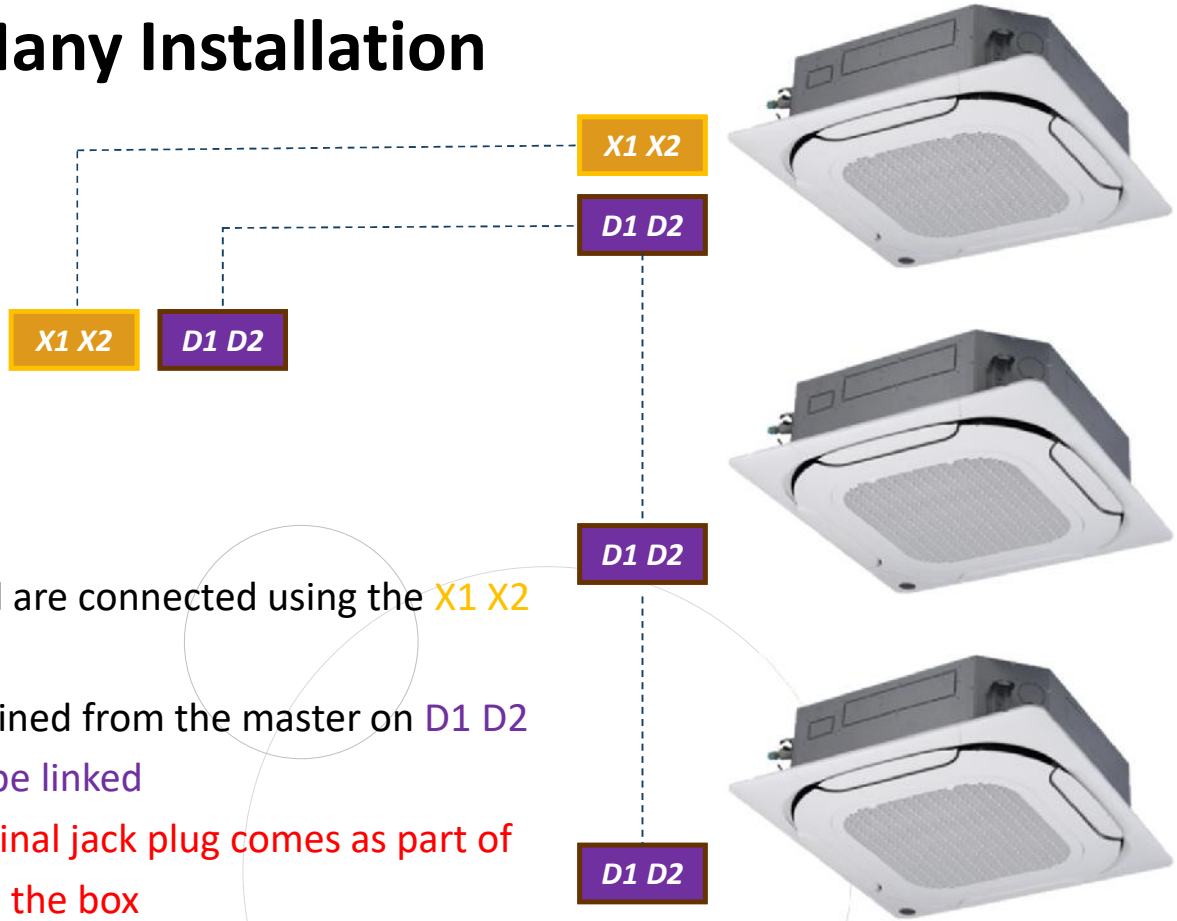
MI2 Unit - 1 to 1 Installation



- Controller and fan coil are connected using the X1 X2 terminals
- 2 core shielded cable 0.5mm²-1.5mm²
- Maximum 200m length

Controller Wiring – WDC120G

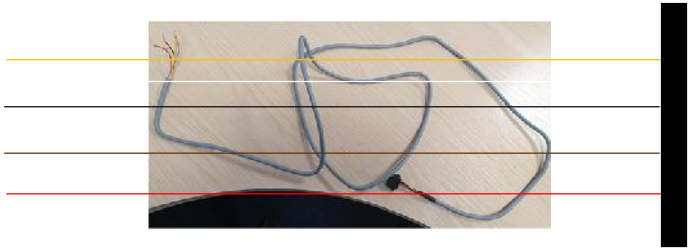
MI2 Unit - 1 to Many Installation



- Controller and master fan coil are connected using the X1 X2 & D1 D2 terminals
- Additional units are daisy chained from the master on D1 D2 terminals. Upto 16 units can be linked
- Please note – the D1 D2 terminal jack plug comes as part of the IDU accessory pack inside the box

Controller Wiring – KJR27B

1 to 1 Installation



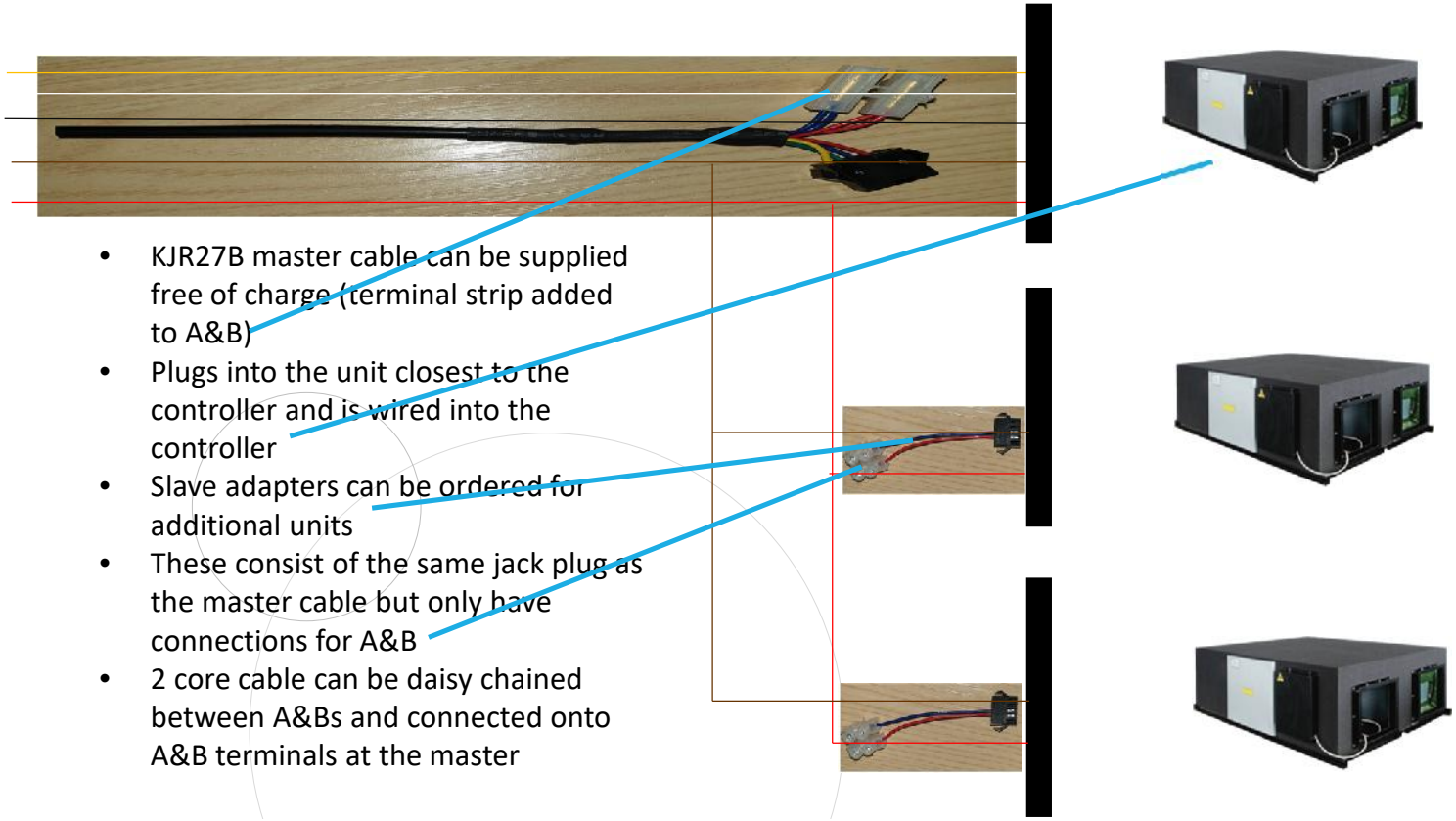
A – Brown/Blue
B – Red
C- Yellow
D – Black
E - White

The installation is very similar to the KJR29B with one extra core (E – white). This provides the 5v power to the LED power indicator

- Comes with a 6m cable
- Jack plugged one side/made off the other
- Jack plugs into ABCDE adapter cable already on the PCB
- Other end is made off and wired into the terminals
- A&B – Comms
- C&D – 5v Power
- E – 5v power for LED

Controller Wiring – KJR27B

1 to Many Installation



- KJR27B master cable can be supplied free of charge (terminal strip added to A&B)
- Plugs into the unit closest to the controller and is wired into the controller
- Slave adapters can be ordered for additional units
- These consist of the same jack plug as the master cable but only have connections for A&B
- 2 core cable can be daisy chained between A&Bs and connected onto A&B terminals at the master

A – Brown/Blue
B – Red
C- Yellow
D – Black
E - White

Pressure Testing, Vacuuming & Charging

- Page 31 Pressure Testing & Vacuuming Heat Pump
- Page 32-33 Pressure Testing & Vacuuming Heat Recovery
- Page 34 Additional Charge – Mini VRF & Sideflow
- Page 35 Additional Charge – V6 Heat Pump
- Page 36 Additional Charge – V4+R Heat Recovery

Pressure Test & Vaccing

Mini VRF, Sideflow & V6

- Test down both pipes at the same time
- Go up slowly in increments of 5bar. Pause every 5bar for an audible check and pipe runs and a visual check of the gauges
- Reach a maximum of 40bar
- Check all mechanical joints with leak spray
- Once you are satisfied there are no leaks leave pressure in for at least 24 hours
- Check pressure after 24 hours. If there is evidence of any leaks, start again. If there are no leaks blow nitrogen off

- Vacuum both pipes at the same time until a reading of 2torr or equivalent is achieved

- Once the desired level is reached add additional charge and open service valves

Pressure Test & Vaccing

V4 + R Heat Recovery

- With any heat recovery system it is imperative to make sure during leak testing and vacuum drying that the full piping circuit is reached. Due to the amount of valves on heat recovery systems and the way these valves act once powered, this can sometimes be tricky to guarantee on other systems. To make this as easy as possible for Midea Heat Recovery systems we have two ways to guarantee that the appropriate valves are open and will allow the full circuit to be reached:

- Power ON all fan coils units in the system. The expansion valves will all set to the standby position of open, allowing the nitrogen/vac to pass the valve and complete the circuit

- Power ON MS Boxes and Condensers. Press SW4 until 000 is shown. Press SW1 (forced cooling button), UA should be displayed and all MS box valves will fully open. All electrically controlled valves at the condensing units will open as well meaning this mode can be used when recovering refrigerant from the system

Pressure Test & Vaccing

V4 + R Heat Recovery

- Test down all 3 pipes at the same time
- Test balance pipes (high gas balance and oil balance) for modular systems individually
- Go up slowly in increments of 5bar. Pause every 5bar for an audible check and pipe runs and a visual check of the gauges
- Reach a maximum of 40bar
- Check all mechanical joints with leak spray
- Once you are satisfied there are no leaks leave pressure in for at least 24 hours
- Check pressure after 24 hours. If there is evidence of any leaks, start again. If there are no leaks blow nitrogen off

- Vacuum both pipes at the same time until a reading of 2torr or equivalent is achieved

- Once the desired level is reached add additional charge and open service valves

Additional Refrigerant Charge

Mini VRF & Sideflow

Calculate the additional charge based on the diameter and length of the liquid line

Liquid Size	Refrigerant per Metre
6.4	0.023kg
9.5	0.060kg
12.7	0.120kg
15.9	0.180kg
19.1	0.270kg
22.2	0.380kg

Additional Refrigerant Charge

V5 & V6 Heat Pump

Calculate the additional charge based on the diameter and length of the liquid line

Liquid Size	Refrigerant per Metre
6.4	0.022kg
9.5	0.057kg
12.7	0.110kg
15.9	0.170kg
19.1	0.260kg
22.2	0.360kg
25.4	0.520kg
28.6	0.680kg

Additional Refrigerant Charge

V4+R Heat Recovery

Calculate the additional charge based on the diameter and length of the liquid line + additional charge for MS Box sizes

Liquid Size	Refrigerant per Metre
6.4	0.023kg
9.5	0.06kg
12.7	0.12kg
15.9	0.180kg
19.1	0.270kg
22.2	0.380kg
25.4	0.550kg
28.6	0.710kg

Model	Additional
MS-02	0.3kg
MS-04/MS-06	0.5kg

Pre-Commissioning Check

- Page 38 FE Error
- Page 39 Commissioning Itinerary

FE Error Code

Prior to being put through a Test Run and/or being addressed, all VRF IDUs will show a **FE** error code. This is completely normal. It simply means they are awaiting an address

Commissioning Itinerary

Once the system is vacuumed, charged and valves open we are ready to set the system up ready for commissioning. The steps are:

- If not already done, power on IDUs and MS Boxes (if applicable)
- IDUs will show FE fault. This is completely normal

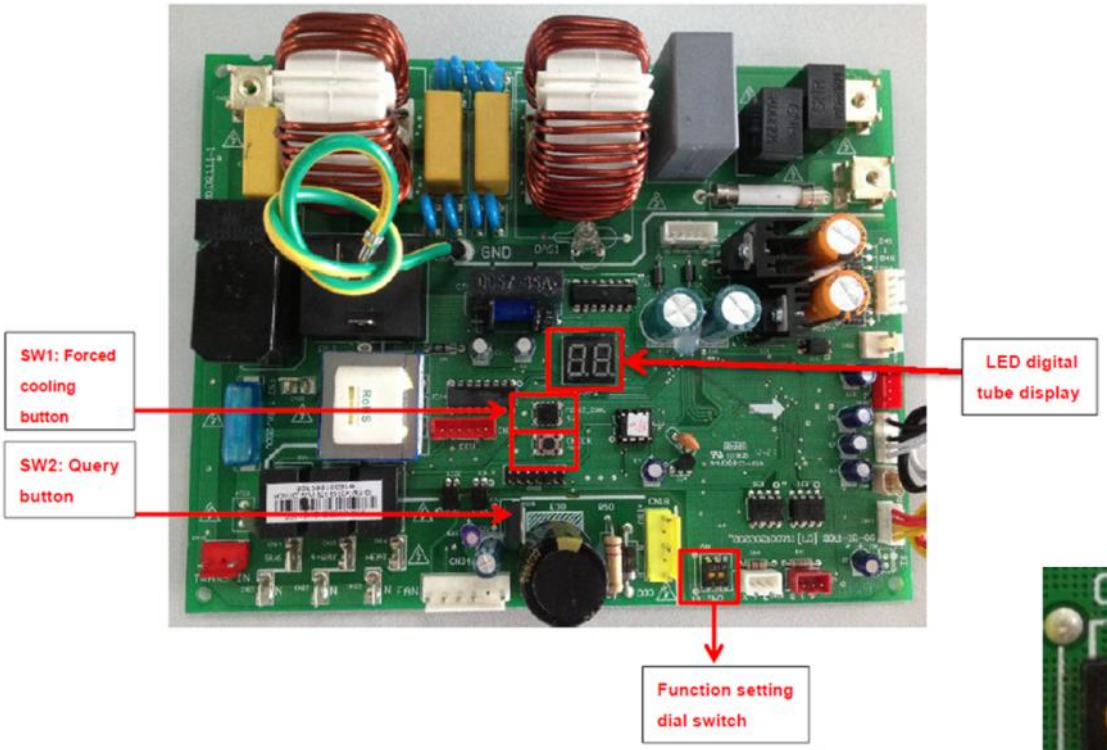
- Set the number of IDUs on the system via the switch on the condensing unit (explained in next chapter)
- Set the system to auto addressing mode using the appropriate dip switch (explained in next chapter). This may already be set depending on the system. This isn't required for V4+R.

- For V4+R Heat Recovery address the IDUs (explained in later chapter). Then Test Run (explained in later chapter)
- For Heat Pump systems simply Test Run the system (explained in later chapter). Addressing is automatic

Auto Addressing Settings

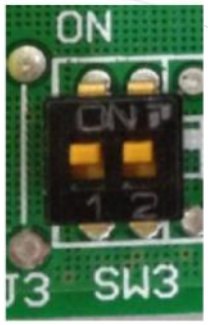
- Page 41 80W/105W Models
- Page 42 120W-160W 1 phase models
- Page 43 120W-180W 3 phase models
- Page 44 200W-450W models
- Page 45 V6

Auto Addressing – 80/105 models



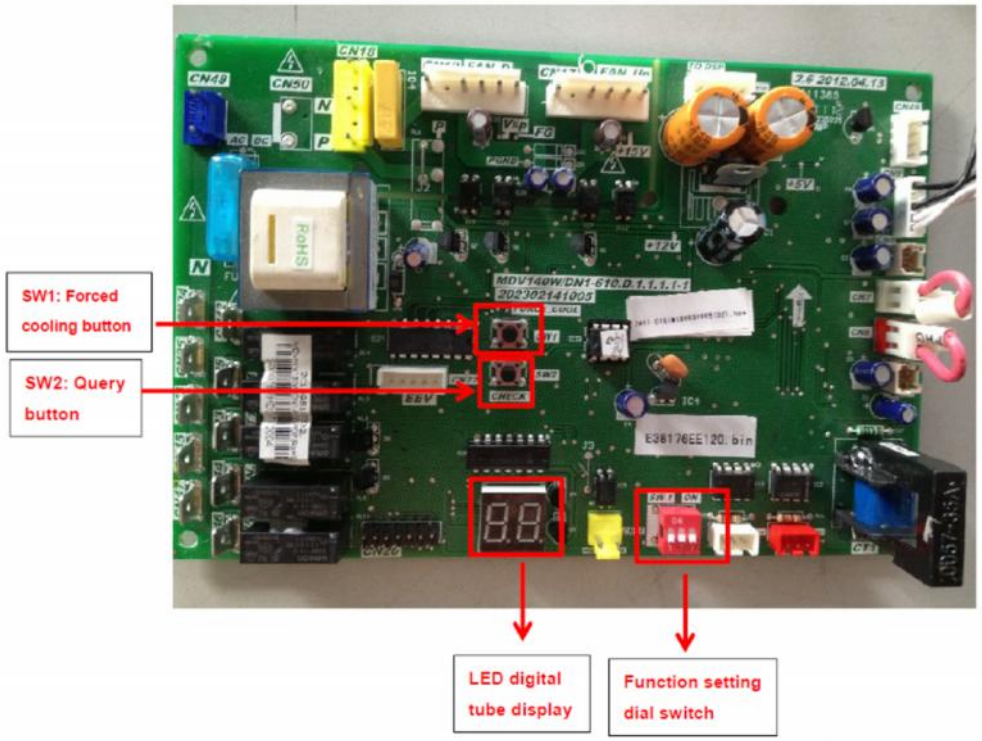
- SW3 is used to set up the auto addressing function at the condensing unit
- VRF Condensers should ALWAYS be set to Auto Address even if you wish to change the addresses manually at a later date
- Enabling auto address allows communication with central controllers

- Bit 1 ON
- Bit 2 OFF



SW3 Settings		
1	ON	Auto Address
	OFF	Manual
2	ON	Clear Addresses
	OFF	Default

Auto Addressing – 120-160 1ph

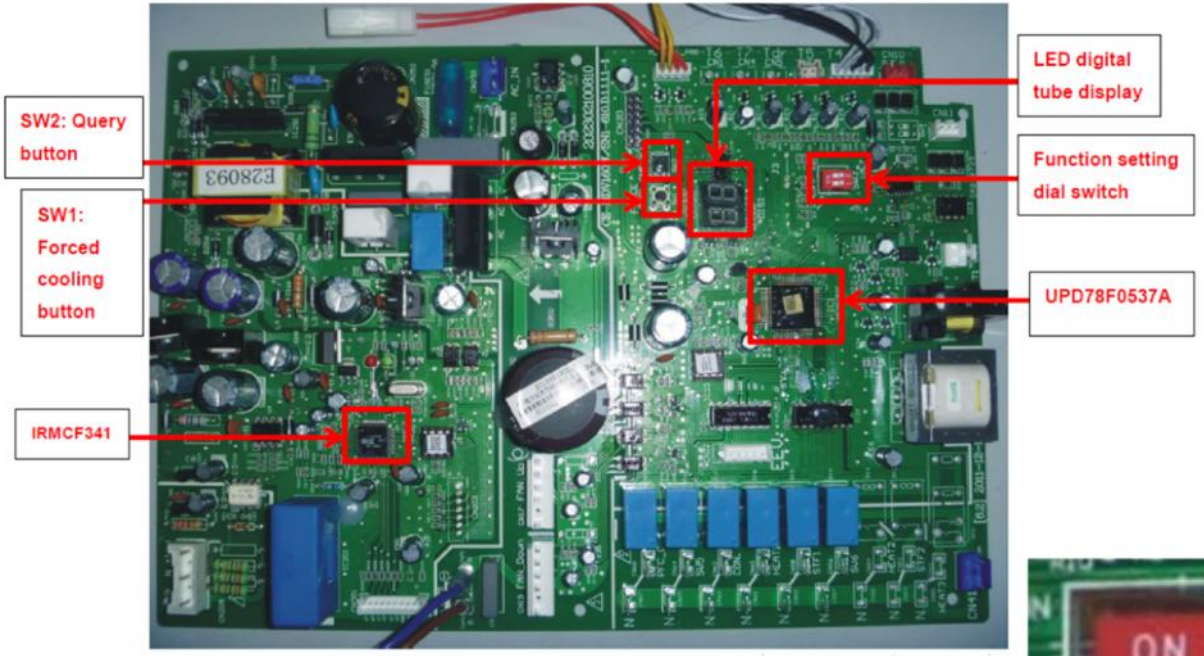


- SW1 is used to set up the auto addressing function at the condensing unit
- VRF Condensers should ALWAYS be set to Auto Address even if you wish to change the addresses manually at a later date
- Enabling auto address allows communication with central controllers

- Bit 1 ON
- Bit 2 OFF
- Bit 3 OFF

SW1 Settings		
1	ON	Auto Address
	OFF	Manual
2	ON	Clear Addresses
	OFF	Default
3	Reserved	Reserved

Auto Addressing – 120-180 3ph



- SW7/SW3 is used to set up the auto addressing function at the condensing unit
- VRF Condensers should ALWAYS be set to Auto Address even if you wish to change the addresses manually at a later date
- Enabling auto address allows communication with central controllers

- Bit 1 ON
- Bit 2 OFF

SW3 Settings		
1	ON	Auto Address
	OFF	Manual
2	ON	Clear Addresses
	OFF	Default

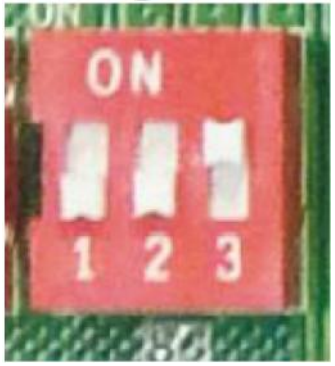
Auto Addressing – 200W-450W

200-260 model

S6: Addressing mode selection	
S6 ON 1 2 3	Auto addressing mode
S6 ON 1 2 3	Manual addressing mode (factory default)
S6 ON 1 2 3	Clean the indoor unit address (only available for auto searching new indoor unit)

400-450 model

S6

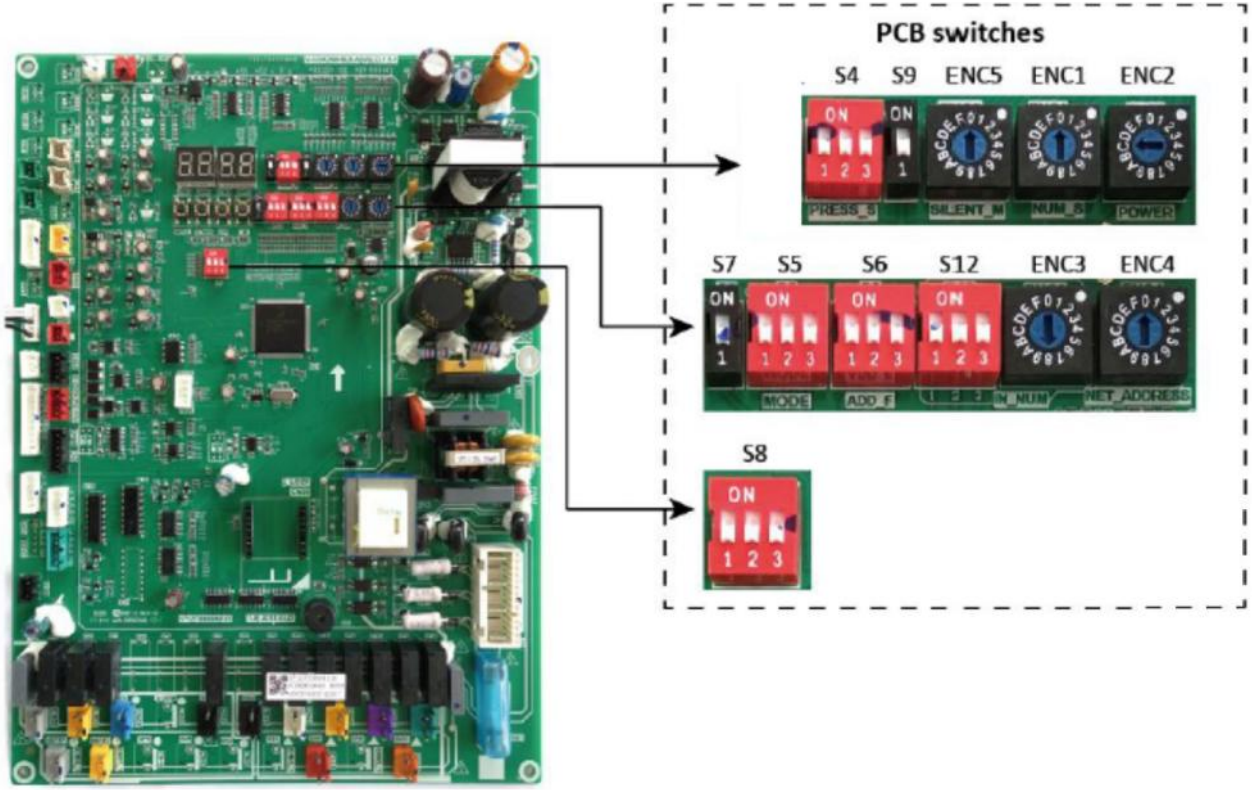


S6: Addressing mode selection	
S6 ON 1 2 3	Auto addressing mode
S6 ON 1 2 3	Manual addressing mode (factory default)
S6 ON 1 2 3	Clean the indoor unit address (only available for auto searching new indoor unit)

PLEASE NOTE – ON THE ILLUSTRATIONS THE BLACK REPRESENTS THE SWITCH HEAD

- Move all Bits on the Dip Switch to the OFF position for auto addressing
- Once Auto addressing is set leave in this position even if you wish to manually change the addresses at a later date

Auto Addressing – V5 & V6



S6 Dip Switch

The Dip switch is factory set to auto address

All Bits to the OFF position

The addresses will automatically be given once the condenser is switched on

The clear addresses switch Bit 2 to ON position, wait 2 mins, power off, switch Bit 2 back to OFF position, switch power back on

Switch	Setting	Switch positions[1]	Description
S6-1	Reserved	ON 1 2 3	Reserved
S6-2	Clear indoor unit addresses	ON 1 2 3	No action (default)
		ON 1 2 3	Clear indoor unit addresses
S6-3	Addressing mode	ON 1 2 3	Auto addressing (default)
		ON 1 2 3	Manual addressing

Condenser Settings

- Page 47-49 Capacity & Outdoor Addressing
- Page 50 Quantity of IDUs
- Page 51-53 Silent Mode
- Page 54 Static Pressure
- Page 55-56 Mode Priority

PCB Settings – ENC2/3/4

Sideflow VRF – Capacity / Outdoor Address

Switches	Setting	Description
ENC2	Outdoor unit capacity	Only 22.4/26kW units have ENC2 dial switch. ENC2 is used for outdoor unit capacity setting (F refer to 22.4kW; 0 refer to 26kW)
ENC3	Outdoor unit network address	Only 22.4/26kW units have ENC3 dial switch. ENC3 is used for outdoor unit address setting (0-F refer to 0-15)

20-26kw

Switches	Setting	Description
ENC1	Reserved	Reserved
ENC2	Outdoor unit capacity	Outdoor unit capacity setting: 4-5 are available (capacity code4—40kW; capacity code 5—45kW)
ENC4	Outdoor unit network address	Outdoor unit network address setting (0-7 is effective)

40-45kw

PCB Settings – ENC1/2/3

V4+R – Capacity / Outdoor Address

Dial Switch	Content	Note
ENC3	System address setting	Address for Central Controller
ENC2	Capacity Setting	0: 8HP, 1: 10HP, 2: 12HP, 3:14HP, 4: 16HP
ENC1	Master/Slave Address	0: Master unit, 1-3: Slave units

PCB Settings – ENC1/2/4

V6 – Capacity / Outdoor Address

Switch	Setting	Switch positions[1]	Description
ENC1	Outdoor unit address		Only 0, 1, 2 should be selected (default is 0) 0 is for master unit; 1 and 2 are for slave units
ENC2	Outdoor unit capacity		Only 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C should be selected 0: 8HP; 1: 10HP; 2: 12HP; 3: 14HP; 4: 16HP; 5: 18HP; 6: 20HP; 7: 22HP; 8: 24HP; 9: 26HP; A: 28HP; B:30HP; C:32HP
ENC4	Network address		Only 0, 1, 2, 3, 4, 5, 6, 7 should be selected (default is 0)

NOTE:

1. ***Black denotes the switch position.***

PCB Settings – ENC3

Quantity of Indoors

Switches	Setting	Switch positions	Description
ENC3 S12	Quantity of indoor units	ENC3 S12 ON	The quantity of indoor unit is 0-15. 0~9 on ENC3 refer to 0~9 indoor units; A~F on ENC3 refer to 10~15 indoor units.
		ENC3 S12 ON	The quantity of indoor unit is 16-31. 0~9 on ENC3 refer to 16~25 indoor units; A~F on ENC3 refer to 26~31 indoor units.
		ENC3 S12 ON	The quantity of indoor unit is 32-47. 0~9 on ENC3 refer to 32~41 indoor units; A~F on ENC3 refer to 42~47 indoor units.
		ENC3 S12 ON	The quantity of indoor unit is 48-63. 0~9 on ENC3 refer to 48~57 indoor units; A~F on ENC3 refer to 58~63 indoor units.

NOTE:

1. ***Black denotes the switch position***

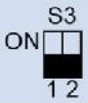
You can set the number of indoor units the system is looking for using ENC3

This is NOT required for sideflow VRF but if it is set it will allow the system to fault out if a fan coil unit is lost due to power failure

This is required for V5/V6 and V4+R Heat recovery

PCB Settings – S3

Silent Mode

Switches	Setting	Switch positions	Description
S3	Silent mode		Nighttime silent mode (factory default)
			Silent mode
			Super silent mode
			No silent mode

You can select the Silent mode using S3

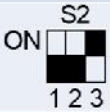
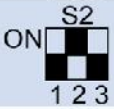
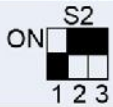
The settings allow you select which silent mode logic, if any, you want the condensers to follow

NOTE:

1. ***Black denotes the switch position***

PCB Settings – S2

Night Silent Time

Switches	Setting	Switch positions	Description
S2	Night silent time		Nighttime silent time 6h/10h (factory default)
			Nighttime silent time 6h/12h
			Nighttime silent time 8h/10h
			Nighttime silent time 8h/12h

You can set the Night silent time using S2

The settings allow you to adjust the time it takes to begin after the peak of the day and for how long it is active

NOTE:

1. ***Black denotes the switch position***

PCB Settings – ENC5 – V6

Silent Mode

Switch	Setting	Switch positions[1]	Description
ENC5	Silent mode	0	Night silent time is 6h/10h (default)
		1	Night silent time is 6h/12h
		2	Night silent time is 8h/10h
		3	Night silent time is 8h/12h
		4	No silent mode
		5	Silent mode 1 (only limit max. fan speed)
		6	Silent mode 2 (only limit max. fan speed)
		7	Silent mode 3 (only limit max. fan speed)
		8	Super silent mode 1 (limit max. fan speed and compressor frequency)
		9	Super silent mode 2 (limit max. fan speed and compressor frequency)
		A	Super silent mode 3 (limit max. fan speed and compressor frequency)
		B	Super silent mode 4 (limit max. fan speed and compressor frequency)
		F	Set silent mode via centralized controller (reserved)

You can select the Silent mode using ENC5

The settings allow you select which silent mode logic, if any, you want the condensers to follow

PCB Settings – S4

Static Pressure Settings (N/A for Mini VRF & Sideflow)

Switch	Setting	Switch positions[1]	Description
S4	Static pressure		Standard static pressure (default)
			Low static pressure mode (reserved)
			Medium static pressure mode (reserved)
			High static pressure mode (reserved)
			Super high static pressure mode (reserved)

You can set the static pressure of the outdoor fans using S4

You can increase the 'throw' of the fan if duct is to be fitted to the cowl

NOTE:

1. ***Black denotes the switch position***

PCB Settings – S5

Mode Priority

Switches	Setting	Switch positions	Description
S5	Priority mode		Heating priority mode(factory default)
			Cooling priority mode
			No. 63 address takes priority
			Only response to heating mode
			Only response to cooling mode

You can set the mode priorities of the system using S5

These priorities will be used as the basis of mode choice in any mode conflicts or auto mode logic issues

NOTE:

1. ***Black denotes the switch position***

PCB Settings – S5 – V6

Mode Priority

Switch	Setting	Switch positions[1]	Description
S5	Priority mode		Auto priority (default)
			Cooling priority
			VIP priority or voting priority
			Heating only
			Cooling only
			Set priority mode via centralized controller (reserved)

You can set the mode priorities of the system using S5

These priorities will be used as the basis of mode choice in any mode conflicts or auto mode logic issues

NOTE:

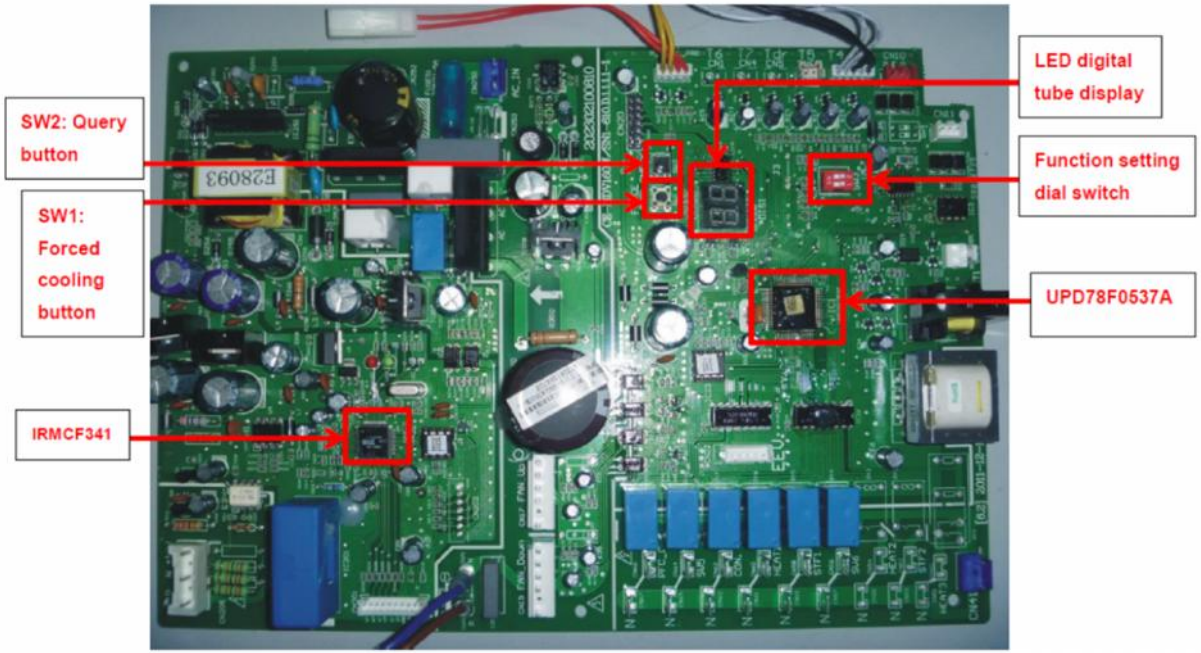
1. ***Black denotes the switch position***

Commissioning

- Page 58 Mini VRF & Sideflow
- Page 59 V4+R Heat Recovery
- Page 60-61 V6 Heat Pump
- Page 62 ODU Digital Display Parameter Checking

Commissioning - Test Running

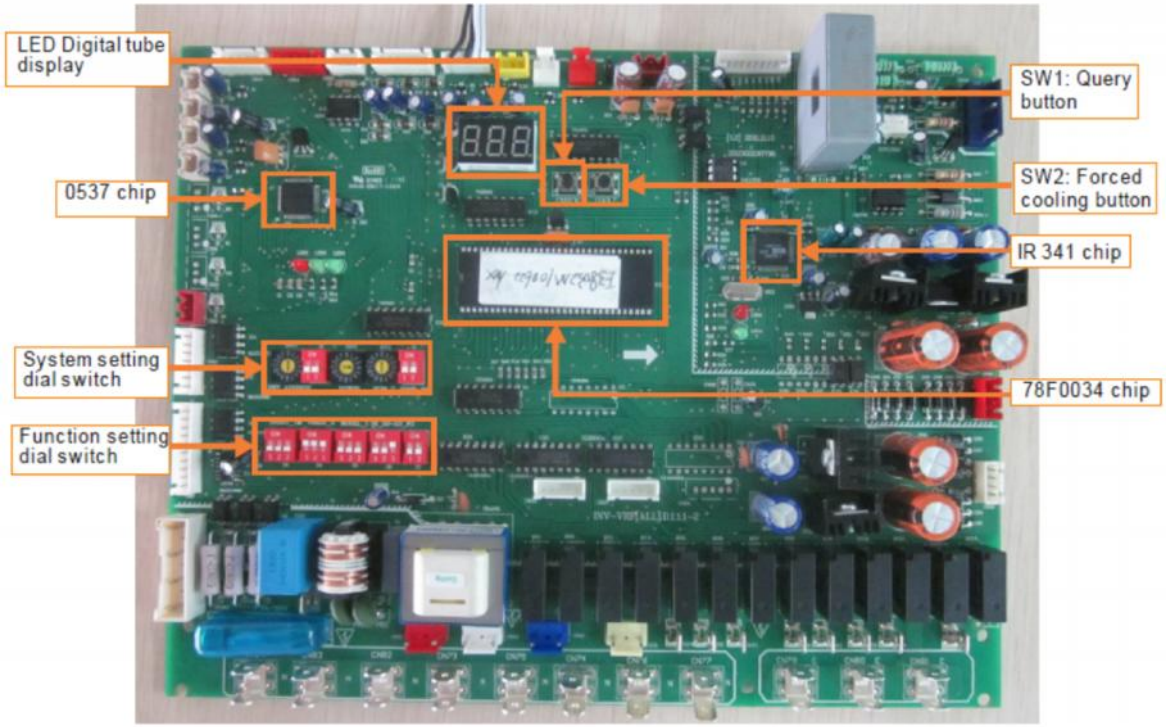
Mini VRF & Sideflow VRF



- Once the dip switches are set, power on the condenser and the fan coils (if not already powered)
- Press SW1 Forced Cooling to send the addresses to the fan coils and initiate a 1hr Test Cooling mode
- Press the button again to end the Test Mode

Commissioning - Test Running

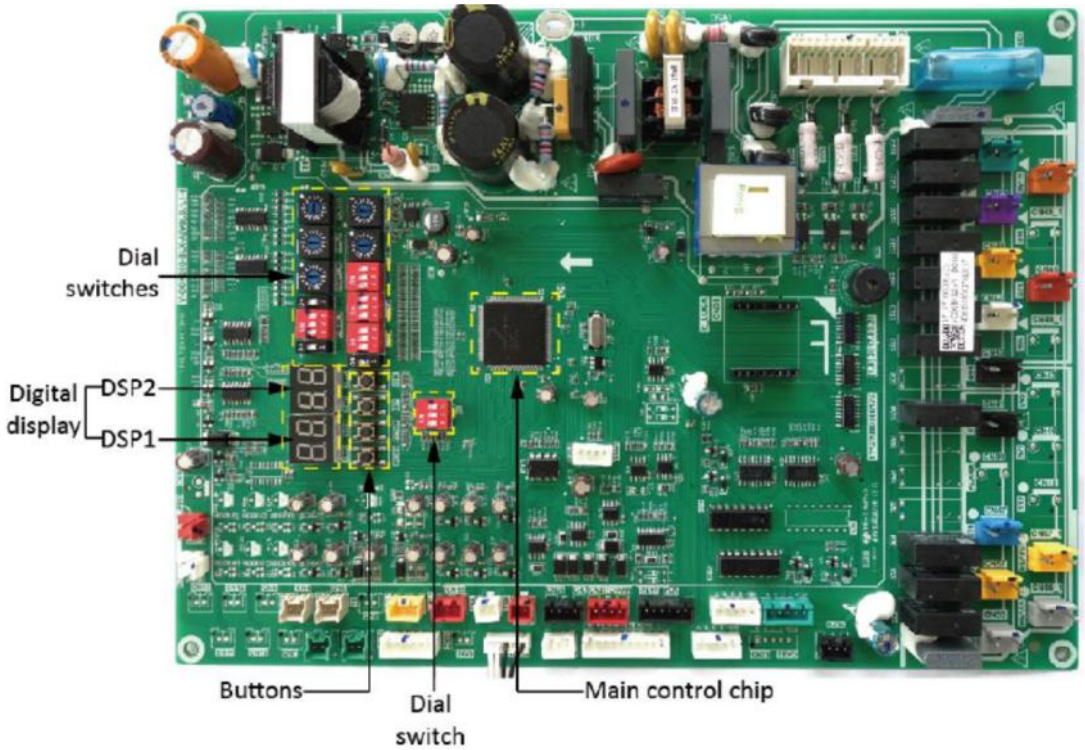
V4+R Heat Recovery

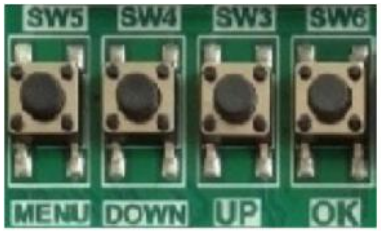


- Once the dip switches are set, power on the condenser and the fan coils (if not already powered)
- Indoor units must now be addressed using any of the methods in the IDU ADDRESSING SECTION of the guide
- Press SW2 Forced Cooling to initiate a 1hr Test Cooling mode
- Press the button again to end the Test Mode

Commissioning - Test Running

V6 Heat Pump





Test function is accessed using the commissioning menu on V6 systems

Button	Function
SW3 (UP)	In the menu mode: previous and next buttons for menu modes Not in menu mode: previous and next buttons for system check information
SW4 (DOWN)	
SW5 (MENU)	Enter/exit menu mode
SW6 (OK)	Confirm

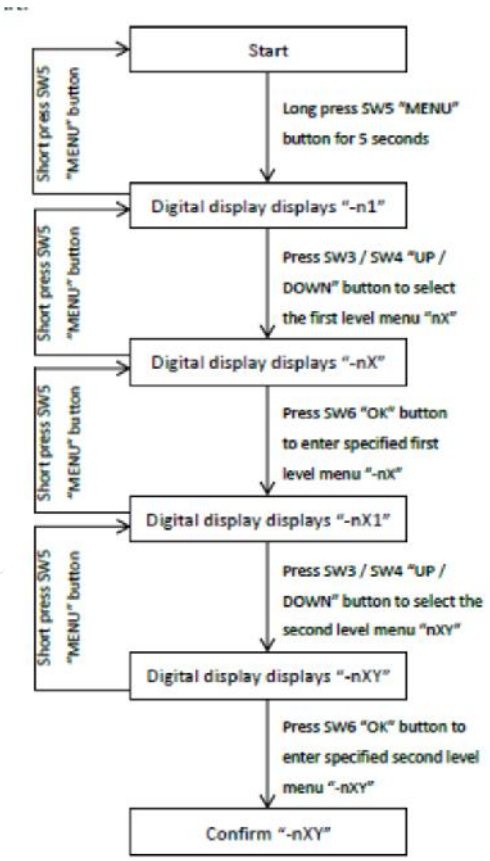
Commissioning - Test Running

V6 Heat Pump

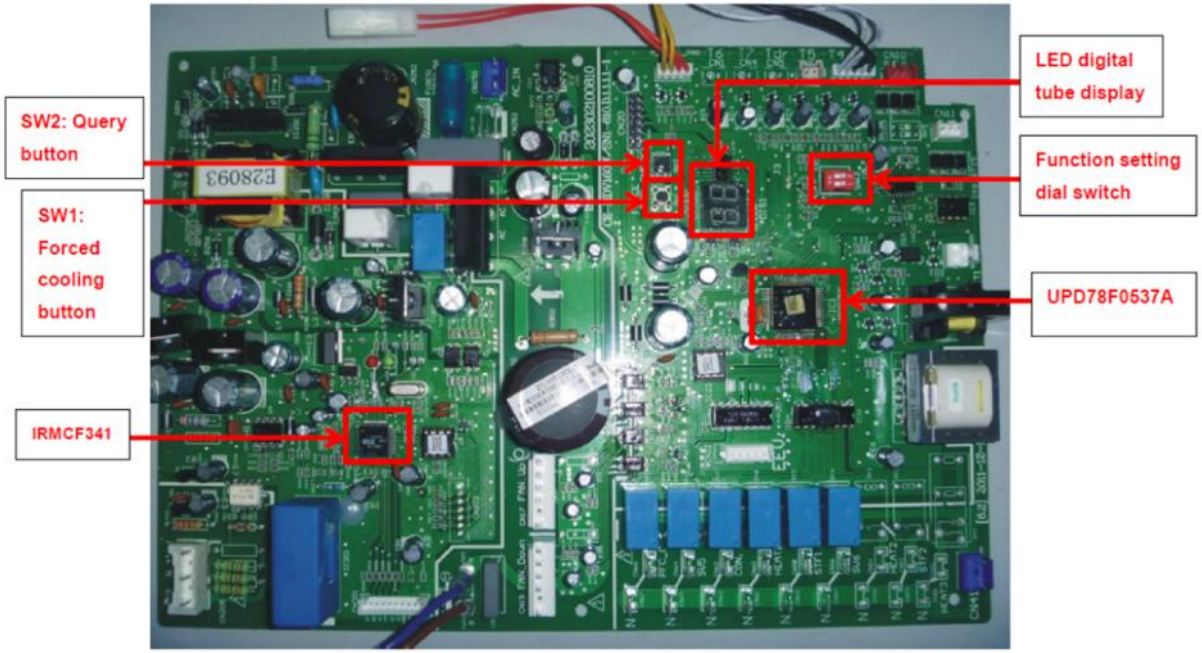
Only the master unit has the full menu functions. Slave units only have error codes and cleaning functions

1. Long press SW5 “Menu” button for 5 seconds. Digital display changes to “n1”
2. Press SW3/SW4 to navigate to the first level menu. n1 is the required menu for test
3. SW6 to select menu
4. Press SW3/SW4 to navigate second level menu “n12”, “n15” etc
5. “n14” is the selection for test cooling

6. Press SW6 to confirm choice



ODU – Digital Display



Using SW1/SW2: Query Button you can get the operating parameters of the system. This will be shown on the LED digital tube display

The table of available data will be shown on the inside of the condenser panel. The items available include thermistor readings, capacity requirements and recent errors

In Standby the quantity of indoor units is displayed

Whilst running the frequency of the compressor is shown

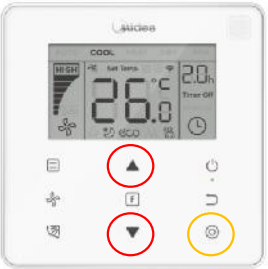
Whilst in error the current Error code is shown

IDU Addressing

- Page 64 Addressing using WDC86E/KD
- Page 65 Addressing using WDC120G
- Page 66 Addressing using RM12 infra red
- Page 67 Addressing using KJR29B
- Page 68 HRV Addressing

IDU Addresses – WDC86E/KD

Auto addressing using SW1 Forced Cooling from the condensing unit will address the IDUs in no particular order. After the Test mode has ended you may wish to know the addresses or change the addresses. We can do this with either a hard wired (when connected to one unit only) or infra red controller:

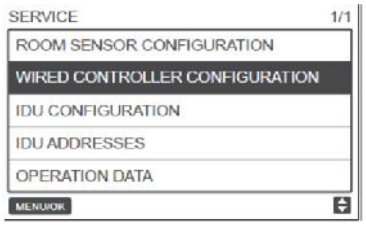


- Press and hold the UP and DOWN buttons for 8 seconds to enter the Addressing menu
- If the IDU has a current address it will be shown on the screen
- To amend the address, use the UP and DOWN buttons to select the required address and press the CONFIRM button to send it

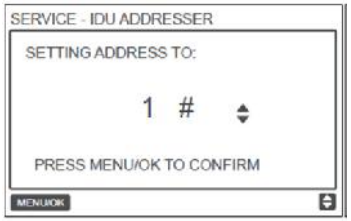
IDU Addresses – WDC120G

Auto addressing using SW1 Forced Cooling from the condensing unit will address the IDUs in no particular order. After the Test mode has ended you may wish to know the addresses or change the addresses. We can do this with either a hard wired (when connected to one unit only) or infra red controller:

- When the backlight of the screen is off, press and hold BACK and FAN at the same time for 5 seconds
- Use UP and DOWN arrows to scroll to IDU ADDRESSES

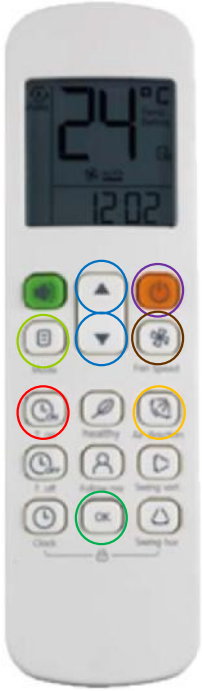


- Press the MENU/OK button
- Use UP and DOWN to select an address and press MENU/OK to send the address to the IDU
- Press BACK twice to exit the parameter settings



IDU Addresses – RM12 Infrared

Auto addressing using SW1 Forced Cooling from the condensing unit will address the IDUs in no particular order. After the Test mode has ended you may wish to know the addresses or change the addresses. We can do this with either a hard wired (when connected to one unit only) or infra red controller:



- Hold TIMER ON, AIR DIRECTION, and OK buttons for 8 seconds to enter Addressing Mode
- Use the UP and DOWN buttons to select an address

- Press POWER button to ready the controller for the transfer

Power indicator should appear

- Press FAN SPEED to send the address to the IDU. It should appear on the IDU display if it has been successful
- Press MODE to interrogate the address. The current address will be shown on the IDU display
- Press OK to leave Addressing Mode

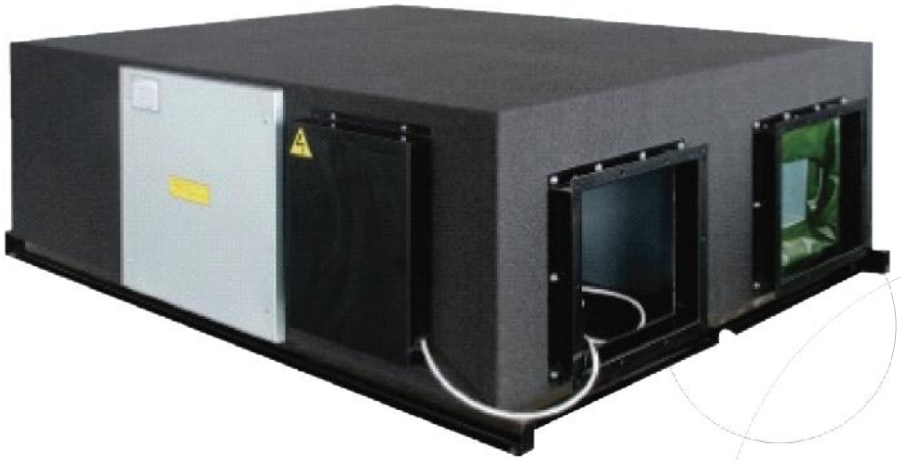
IDU Addresses – KJR29B

Auto addressing using SW1 Forced Cooling from the condensing unit will address the IDUs in no particular order. After the Test mode has ended you may wish to know the addresses or change the addresses. We can do this with either a hard wired (when connected to one unit only) or infra red controller:

- Press and hold the UP and DOWN buttons for 8 seconds to enter the Addressing menu
- If the IDU has a current address it will be shown on the screen
- To amend the address, use the UP and DOWN buttons to select the required address and press the Fan Speed button to send it
- Press OK to leave the setting

IDU Addresses – HRV units

Auto addressing using SW1 Forced Cooling from the condensing unit will address the IDUs in no particular order. After the Test mode has ended you may wish to know the addresses or change the addresses. We can do this with either a hard wired (when connected to one unit only) or infra red controller:



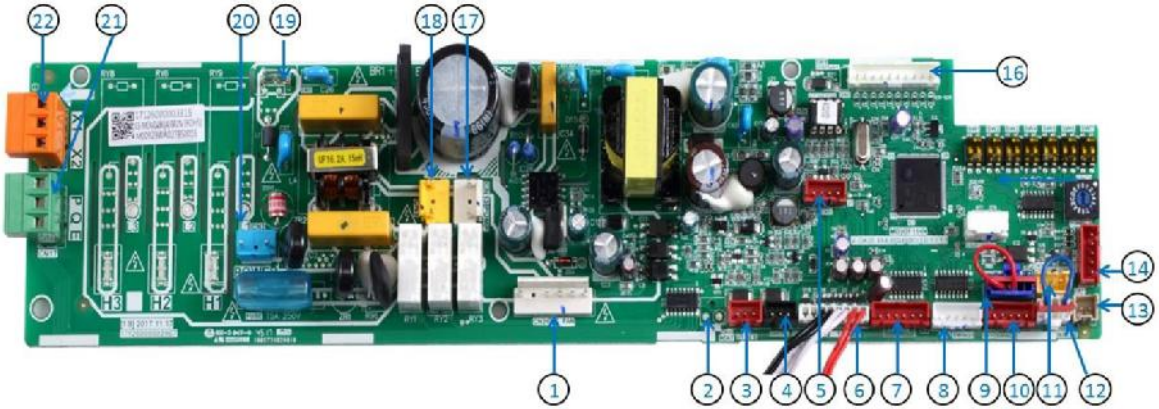
HRV units can be auto addressed as any other IDU

They can also be addressed using KJR29B and infra red controller. The PCB of HRV has inbuilt infra red receiver to accept address from RM12 controller

IDU/MS Box Connections & Set up

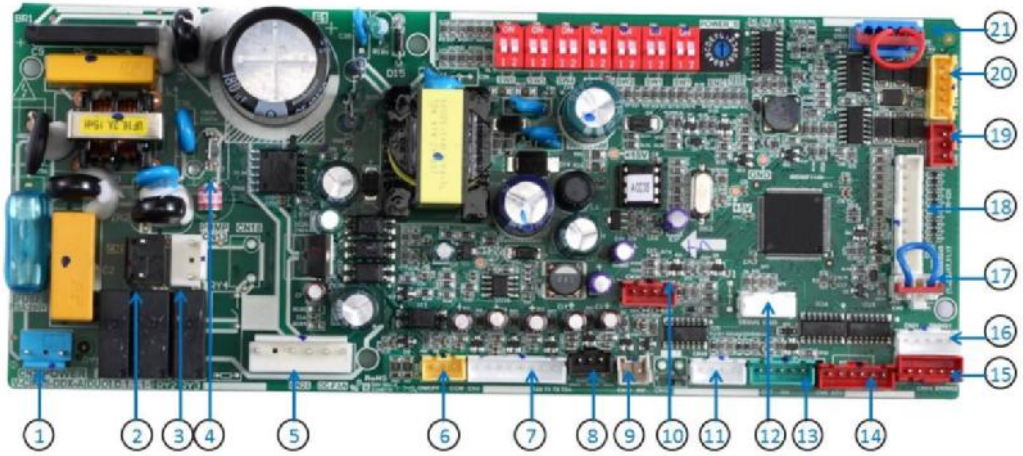
- Page 70 Cassette PCB Connections
- Page 71 Compact PCB Connections
- Page 72 Ducted PCB Connections
- Page 73 Ceiling Floor PCB Connections
- Page 74 Wall mounted Connections
- Page 75 Cooling mode Temp. compensation
- Page 76 Heating mode Temp. compensation
- Page 77 Heating mode fan cycle
- Page 78 Cold Draft Preventation
- Page 79 Auto Restart
- Page 80 Static Pressure
- Page 81-82 Digital Displays
- Page 83-84 Parameter Checking

IDU Connections - Cassette



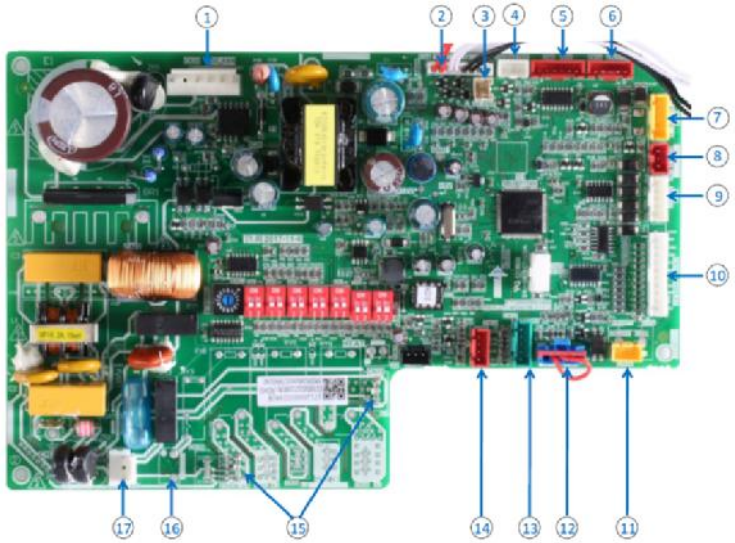
Label	Code	Content	Voltage
22	CN17	PQE Comms port	2.5v-2.7v DC
21	CN18	X1 X2 Comms port	18v DC
3	CN9	D1 D2 Comms port	2.5v-2.7v DC
16	CN15	Display Panel Connection	5v DC
8	CN21	Vertical Louver	12v DC

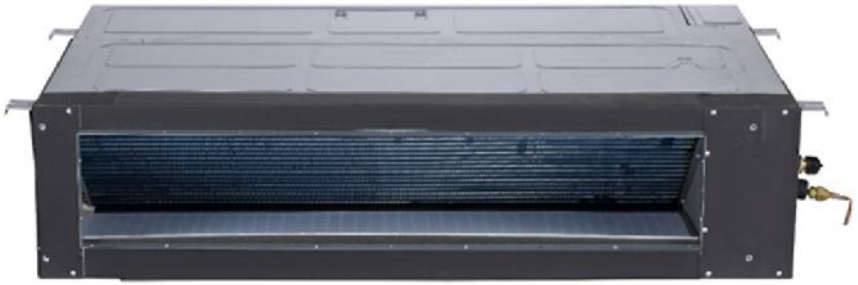
IDU Connections - Compact



Label	Code	Content	Voltage
20	CN17	PQE Comms port	2.5v-2.7v DC
20	CN17	X1 X2 Comms port	18v DC
19	CN9	D1 D2 Comms port	2.5v-2.7v DC
18	CN15	Display Panel Connection	5v DC
16	CN21	Vertical Louver	12v DC

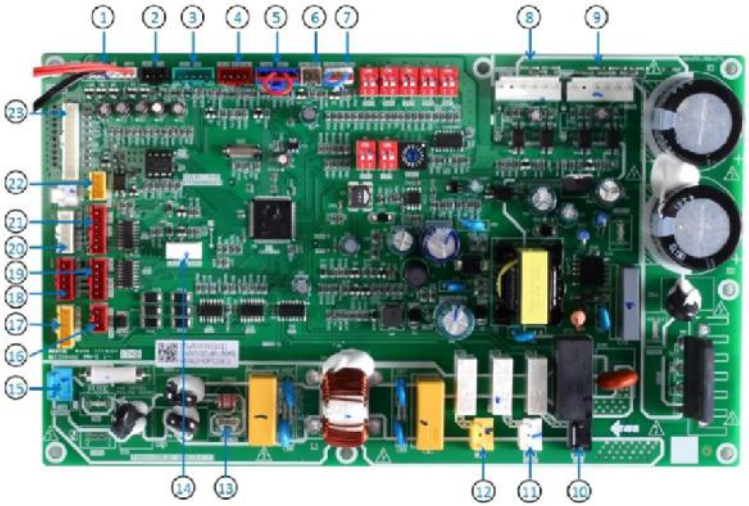
IDU Connections - Ducted





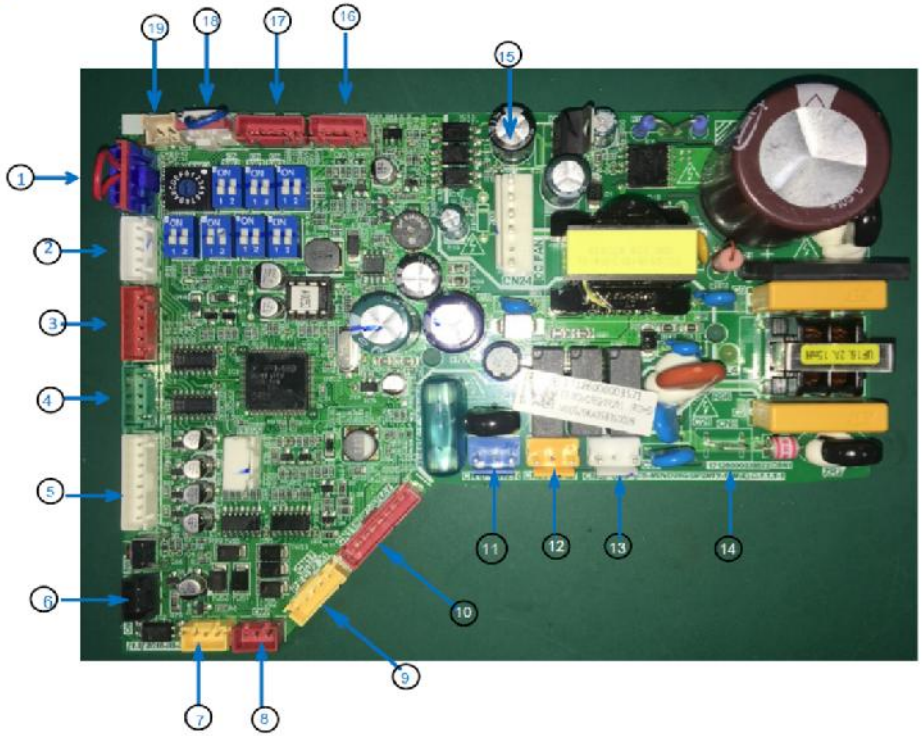
Label	Code	Content	Voltage
7	CN17	PQE Comms port	2.5v-2.7v DC
7	CN17	X1 X2 Comms port	18v DC
8	CN9	D1 D2 Comms port	2.5v-2.7v DC
10	CN15	Display Panel Connection	5v DC

IDU Connections – Ceiling Floor



Label	Code	Content	Voltage
17	CN17	PQE Comms port	2.5v-2.7v DC
17	CN18	X1 X2 Comms port	18v DC
16	CN9	D1 D2 Comms port	2.5v-2.7v DC
23	CN15	Display Panel Connection	5v DC
18/19 /20	CN21/CN22/CN14	Vertical Louver	12v DC

IDU Connections – Wall Mounted



Label	Code	Content	Voltage
9	CN17	PQE Comms port	2.5v-2.7v DC
9	CN17	X1 X2 Comms port	18v DC
8	CN9	D1 D2 Comms port	2.5v-2.7v DC
10	CN11	Display Panel Connection	5v DC
2	CN13	Vertical Louver	12v DC

IDU Field Settings

Cooling Mode Temperature Compensation

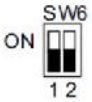
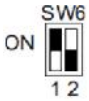
Switch Example	Setting	Switch Positions	Description
SW1_1	Cooling Mode Compensation		0°C (default)
			2°C

Using this setting the indoor unit can target a temperature that is lower than the set temperature

For example a setpoint of 21°C with the compensation of 2°C would target a set point of 19°C (as sensed at the unit)

IDU Field Settings

Heating Mode Temperature Compensation

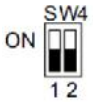
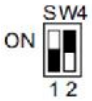
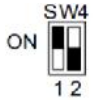
Switch Example	Setting	Switch Positions	Description
SW6	Heating Mode Temperature Compensation		Temperature compensation is 6°C (default)
			Temperature compensation is 2°C
			Temperature compensation is 4°C
			Temperature compensation is 0°C

Since warm air rises and units are often installed at ceiling height, the sensed temperature at the unit can be higher than that of the actual reading at ground level. We can compensate for this using this field setting

A set point of 24°C with the default field setting will aim for a unit return air temp of 30°C

IDU Field Settings

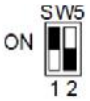
Heating Mode Fan Cycle

Switch Example	Setting	Switch Positions	Description
SW4	Heating Mode Fan Cycle		Set point reached, fan cycles 4 mins OFF, 1 min ON (default)
			Set point reached, fan cycles 8 mins OFF, 1 min ON
			Set point reached, fan cycles 12 mins OFF, 1 min ON
			Set point reached, fan cycles 16 mins OFF, 1 min ON

The logic of the fan cycle once the set point has been reached in heating mode can be changed to suit the end users preference

IDU Field Settings

Cold Draft Prevention

Switch Example	Setting	Switch Positions	Description
SW5	Cold Draft Prevention		Fan does not run until coil temp is 15°c (default)
			Fan does not run until coil temp is 20°c
			Fan does not run until coil temp is 24°c
			Fan does not run until coil temp is 26°c

In heating mode you can select at what coil temperature you would like the fan to begin cycling. This allows you to prevent cold air being recirculated after defrost cycles

IDU Field Settings

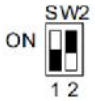
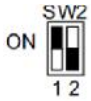
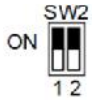
Auto Restart

Switch Example	Setting	Switch Positions	Description
J1	Auto Restart		Enabled (default)
			Disabled

This function can ensure that, in the event of a power outage, the indoor units automatically restart once the power returns. The restart of the units is staggered to prevent all units starting simultaneously

IDU Field Settings

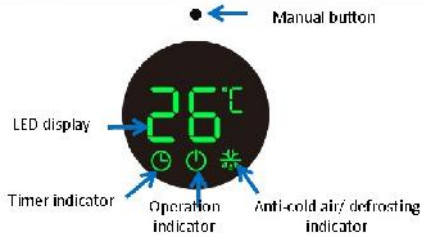
Static Pressure

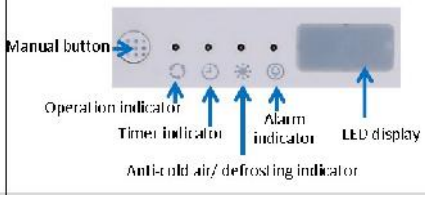
Switch Example	Setting	Switch Positions	Description
SW2	Static Pressure		Mode 1(default)
			Mode 2
			Mode 3
			Mode 4

Cap.	Mode 1	Mode 2	Mode 3	Mode 4
2.2-7.1kw	10pa	0pa	30pa	50pa
8-11.2kw	20pa	40pa	70pa	100pa
14kw	40pa	70pa	100pa	150pa

The static pressures can also be changed using the WDC120G hard wired controller, this time with 9 separate settings

IDU Display – Appearance

Display panel for four-way cassette (Standard panel)	Display panel for four-way cassette (New 360 degree panel)
	

Display panel for compact four-way cassette	Display panel for one-way cassette
	

Display panel for ceiling and floor	Display panel for two-way cassette, floor standing, medium static pressure duct and high static pressure duct

All VRF IDUs come with a digital display that shows information on the current running status and mode

These displays also allow easy diagnose of error codes and installation issues

The ‘Manual’ button can be used during commissioning to query the current running parameters of the unit

IDU Display– Normal Output

Unit State		Display Output	
		Light/Icons	Digital Display
Standby		OPERATION light flashes slowly	
Auto Restart Delay			
Shutting-down		All lights off	
Operating	Normal Operation	OPERATION light/icon on	Cool/Heat mode – Set Temp
			Fan only mode - Ambient
	Cold draft prevention/Defrost	OPERATION light/icon on DEF/FAN	Set Temperature
A timer has been set		TIMER light/icon on	N/A

IDU Display – Parameter Output

Parameter No.	Parameter	Remarks
1	Communication Address	0-63
2	Capacity as set on switch ENC1	Unit: HP
3	Network Address	0-63
4	Set Temperature	
5	Indoor Ambient Temperature	Min Value: - 9°C
6	RESERVED	Min Value: - 9°C
7	Indoor heat exchanger mid-point temperature	Min Value: - 9°C
8	Indoor heat exchanger outlet temperature	Min Value: - 9°C
9	Most Recent Error code	“E-” is displayed in no error has occurred since the unit was powered on
10	-	-

By pressing the manual button you can cycle through the parameters in the table

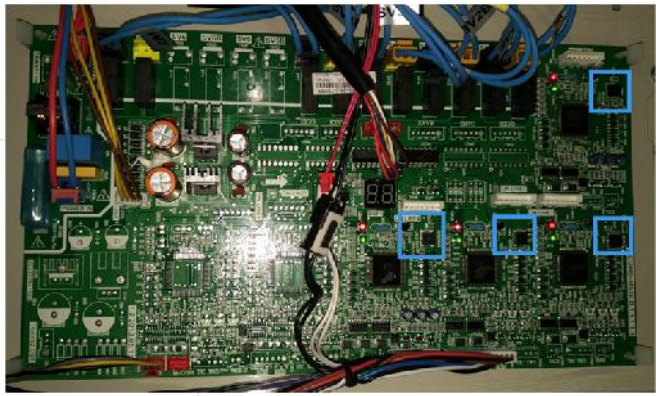
Communications address and Network address are the same value

MS Box Display – Parameter Output

Parameter No.	Parameter
1	Quantity of indoor units connected to this port
2	Operation Mode of this MS
3	Subcooling Inlet Temperature (T1C2)
4	Subcooling Outlet Temperature (T1C1)
5	Total T2(B) average back from outdoor
6	Indoor unit T2(B) value under MS Box
7	Quantity of opened outdoor unit
8	Operation mode of outdoor unit
9	Subcooling valve opening degree
10	Quantity of opened indoors of this MS Box

By pressing the check button you can cycle through the parameters in the table

Each PQE wiring terminal/port has its own check button (SW1-SW6)



0: OFF, 2: Cooling, 3: Heating, 4: Forced Cooling, 5: Main Cooling, 6: Main Heating